

# Predictive Retail Inventory Optimization System

A robust, end-to-end Python pipeline that leverages machine learning to forecast retail demand and dynamically optimize inventory levels. By integrating time-series preprocessing, predictive modeling, and automated inventory replenishment logic (safety stock and reorder points), this system minimizes both stockouts and excess holding costs.

## Features

- **Data Preprocessing & Feature Engineering:** Handles missing values, structures retail sales data for time-series analysis, and extracts temporal features (day of week, month, seasonality).
- **Machine Learning Forecasting:** Evaluates and trains predictive models to forecast future product demand based on historical trends.
- **Dynamic Inventory Logic:** Automatically calculates safety stock and optimal reorder points using predicted demand, lead time, and service level targets.
- **Simulation Environment:** Simulates operations over extended periods to validate model stability and inventory efficiency under varying market conditions.

## System Architecture

The system operates in a sequential, integrated pipeline:

1. **Data Ingestion & Cleaning:** Standardizes input sales historical records.
2. **ML Modeling Phase:** Generates future demand predictions.
3. **Optimization Engine:** Translates forecasts into actionable replenishment signals.
4. **Logging & Audit Framework:** Ensures full system visibility and operational tracking.

## Getting Started

### Prerequisites

```
python >= 3.8  
pandas  
numpy  
scikit-learn
```

## Usage

```
# Clone the repository
git clone
https://github.com/GarrettMeta/predictive-inventory-system.git

# Run the complete pipeline
python main.py
```

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